Multimodal Speaker Identification on BioVid

Extending CNNMC with Visual and Fusion Modalities
under a Lightweight Bayesian Framework

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Abstract

Speaker identification from short audiovisual recordings is an important biometric task with applications in authentication, forensic analysis, and human–computer interaction. This report extends the CNNMC framework proposed by Spata et al. [1] —a convolutional neural network with Monte Carlo dropout for speaker recognition on the BioVid dataset—to the visual and multimodal domains.

Starting from the published audio-only baseline, we design three additional experimental configurations: a compact lightweight audio model based on Global Average Pooling, a lightweight video model operating on uniformly sampled lip frames, and a multimodal late-fusion model combining both modalities. All four systems share a common 5-fold stratified cross-validation protocol, Adam optimization, and Monte Carlo Dropout inference with $T = 30$ stochastic forward passes, exactly matching the evaluation conditions of the reference paper.

Our results reveal a clear performance hierarchy. The lightweight audio model reaches 28.5% MC accuracy (EER = 10.6%), confirming that audio alone is challenging to exploit under tight capacity constraints. Replicating the Spata et al. architecture on audio yields 81.8% MC accuracy (EER = 1.6%), closely approaching the reported benchmark. The lightweight video model surpasses both audio systems with 87.5% MC accuracy (EER = 0.39%) despite using 163× fewer parameters than the paper architecture, establishing a striking *efficiency paradox*: modality quality can dominate model capacity. The multimodal fusion model achieves the best overall result of 94.0% MC accuracy with an EER of 0.19%, confirming that the two modalities contribute complementary identity cues.

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1 Introduction

Speaker identification is the problem of assigning an input recording to one of N known identities, formally a closed-set multi-class classification task. It differs from speaker *verification*, where a binary accept/reject decision is made against a claimed identity. In this work the problem involves 43 speakers, making it a 43-class classification.

The recent work of Spata et al. [1] proposed CNNMC: a CNN-based speaker recognition system on the BioVid dataset that incorporates Monte Carlo (MC) dropout [2] both as a regularizer during training and as a Bayesian inference mechanism at test time. By performing N stochastic forward passes with dropout active, the system obtains improved predictive calibration and explicit uncertainty estimates without requiring a full Bayesian treatment. Their model achieves 93.27% peak validation accuracy and an EER of 0.030 on BioVid when processing audio Mel-spectrograms.

The present work is motivated by a natural extension of that framework to the visual and multimodal domains. The BioVid dataset contains synchronized video recordings in addition to audio, and the video clips provide an additional source of identity-relevant information: the dynamics of lip movements and facial articulation during speech. The first research question is therefore whether the visual stream provides stronger, weaker, or complementary identity cues compared with the audio stream.

A second motivation concerns computational efficiency. The Spata et al. architecture contains approximately 25 million trainable parameters, making it a reasonably large model for the dataset size. We therefore investigate whether a compact architecture—containing roughly 153K parameters and using Global Average Pooling instead of a large dense stage—can achieve competitive performance if the input modality is sufficiently informative. This leads to the second research question: how important is model capacity when modality quality varies?

To address these questions we run four experiments. The first two fix the modality to audio and compare the lightweight and paper architectures. The third and fourth experiments switch to video and multimodal inputs with the lightweight design, allowing a direct assessment of modality contribution under comparable capacity.

2 Related Work

CNNMC (Spata et al., 2025). The reference paper [1] introduces a CNN with Monte Carlo dropout for speaker recognition on the BioVid dataset. The architecture consists of four convolutional blocks followed by a flatten operation, a 512-unit dense layer, and a softmax classifier. Input spectrograms are resized to 128×128 pixels and passed through the network; dropout (rate 0.2) is inserted both after spatial convolutional blocks and inside the classification head. At test time the model performs N forward passes with dropout active and averages the resulting probability vectors, approximating Bayesian inference. The model achieves 93.27% identification accuracy and 0.030 EER on BioVid, outperforming GMM, LDA, X-Vector, I-Vector, and KNN baselines on the same dataset.

Monte Carlo Dropout. Gal and Ghahramani [2] showed that a neural network trained with dropout can be interpreted as a variational approximation to a deep Gaussian process. Keeping dropout active during inference and averaging predictions over T stochastic passes provides both improved accuracy and an estimate of predictive (epistemic) uncertainty without modifying the

training procedure. The uncertainty of a sample is typically measured as the variance of the predicted class probabilities across passes.

Multimodal speaker recognition. Prior work has demonstrated that the visual stream—particularly lip movements—provides speaker-specific cues that complement acoustic features [4]. Late fusion, in which separate audio and video encoders produce embeddings that are concatenated before a joint classifier, is a practical and computationally efficient integration strategy [5].

3 Dataset

The BioVid dataset [1] is a collection of short audiovisual recordings acquired *in the wild* using heterogeneous smartphone microphones, without studio conditions or noise control. It contains 650 video samples from 43 distinct speakers (approximately 15.1 videos per speaker on average), covering both correctly pronounced and incorrectly pronounced password words. Each sample provides a synchronized audio track and a video clip in which the face region has already been pre-cropped to the lip area at a source resolution of 256×128 px (2: 1 width-to-height ratio). Each sample is labeled with the integer identity of the corresponding speaker, yielding a 43-class identification task.

Table 1 summarizes the dataset statistics.

Table 1: BioVid dataset statistics.

Property	Value
Total samples	650
Unique speakers	43
Avg. samples per speaker	15.1
Correctly pronounced	367
Incorrectly pronounced	283
Video resolution (source)	256×128 px (lip region)
Acquisition	In-the-wild, heterogeneous smartphones

Cross-validation protocol. Because the dataset is relatively small, a single train/test split would produce unreliable performance estimates. We adopt stratified 5-fold cross-validation, ensuring that each speaker is represented proportionally in every fold. In each iteration, 80% of the data (approximately 520 samples) is used for training and 20% (approximately 130 samples) for validation; every fold serves once as the held-out set. All reported metrics are means and standard deviations over the five folds. No sample from the same speaker ever appears in both the training and validation sets within any fold.

4 Feature Extraction

4.1 Audio: Mel-Spectrogram with SpecAugment

The raw audio waveform is converted to a Mel-Spectrogram before being fed to any network. A Mel-Spectrogram is a two-dimensional time–frequency representation in which the horizontal axis encodes time frames, the vertical axis encodes frequency on the Mel scale, and each cell encodes the log-energy of the signal. Consistent with Spata et al., we use 128 Mel bins; the same

setting was found by the authors to consistently yield the best accuracy–EER trade-off across their evaluation folds. The resulting representation captures speaker-specific formant patterns, resonance characteristics, and prosodic structure in a form amenable to convolutional processing.

For training, SpecAugment [3] is applied on the fly: random frequency bands and random time steps are masked to zero, creating a different distorted view of the spectrogram at each epoch. SpecAugment operates on the cached tensor in memory without generating additional audio files; it provides regularization through stochastic view diversity. No augmentation is applied to the validation set.

4.2 Video: Uniform Frame Sampling

Each video is represented by a sequence of $N_f = 16$ frames sampled uniformly across the full clip duration. Frames are converted to grayscale and resized to 128×64 px, preserving the 2: 1 source aspect ratio of the lip crop. The frame tensor has shape $(N_f, 1, 64, 128)$ and pixel values are normalized to $[0, 1]$. Short clips containing fewer than 16 frames are tiled to reach the required length. No temporal augmentation is applied to video during training.

The motivation for 16 frames is efficiency: a compact model processes a sparse but representative snapshot of the lip articulation dynamics rather than the full frame sequence, keeping memory footprint low while preserving the temporal structure of mouth movements.

5 Model Architectures

5.1 Lightweight Architecture (GAP)

The lightweight model consists of three convolutional blocks, each containing a convolutional layer (increasing channel counts $32 \to 64 \to 128$), batch normalization, ReLU activation, 2×2 max pooling, and spatial dropout. After the three blocks, **Global Average Pooling** (GAP) replaces each feature map with its spatial mean, producing a 128-dimensional embedding vector. A single linear layer with 43 outputs and softmax activation serves as the classifier.

GAP eliminates the large flatten-and-dense bottleneck found in traditional CNNs, acting simultaneously as a spatial aggregator and a regularizer. This design reduces the total parameter count to approximately **153K**, making it suitable for both audio spectrograms and sequences of video frames.

For the **video** variant, the same CNN encoder is applied frame-by-frame to the $(N_f, 1, 64, 128)$ input, yielding N_f embeddings that are then averaged before the classifier head.

5.2 Paper Architecture (Spata et al. Replication)

To study the effect of architectural capacity while keeping the modality fixed, we replicate the CNNMC architecture of Spata et al. [1]. This model consists of four convolutional blocks followed by a `Flatten` operation and a 512-unit dense layer before the final softmax classifier. Input spectrograms are resized to 128×128 px via bilinear interpolation. Dropout (rate $p = 0.2$) is applied both after the last convolutional block (spatial dropout) and in the dense classification head. The total number of trainable parameters is approximately **25 million**, roughly 163× larger than the lightweight model.

This architecture is used *exclusively in the audio setting* to provide a direct capacity comparison while isolating modality effects.

5.3 Multimodal Late Fusion

The multimodal model uses two independent lightweight encoders—one for audio and one for video—whose 128-dimensional output embeddings are concatenated into a 256-dimensional joint representation. A single 43-class linear layer then performs classification over this concatenated vector. Late fusion is appropriate here because it allows each modality to develop specialized representations before integration, while keeping the combined system computationally light. The total parameter count is approximately **307K**: roughly twice the single-modality lightweight model, and still 80× smaller than the paper architecture.

6 Monte Carlo Dropout

Standard inference in a trained neural network is deterministic: the same input always produces the same output. In the CNNMC framework, dropout layers are left *active at test time*, converting the network into a stochastic function. For each validation sample, $T = 30$ independent forward passes are performed with different random dropout masks. Each pass produces a probability vector $\hat{p}_t \in \mathbb{R}^{43}$, and the final prediction is the class with the highest mean probability:

$$\hat{y} = \arg\max_k \frac{1}{T} \sum_{t=1}^{T} \hat{p}_{t,k}. \tag{1}$$

Predictive uncertainty is estimated as the per-sample standard deviation of class probabilities across the T passes:

$$\sigma_k = \sqrt{\frac{1}{T-1} \sum_{t=1}^{T} (\hat{p}_{t,k} - \bar{p}_k)^2}. \tag{2}$$

MC inference serves two complementary roles. First, it acts as a free ensemble: averaging T diverse predictions smooths out the high-variance outputs that arise from small batch size or low-confidence decisions, consistently improving accuracy over a single deterministic pass. Second, it provides calibrated uncertainty estimates that are especially valuable in biometric contexts, where knowing when *not* to trust a prediction is as important as accuracy.

7 Experimental Setup

All models are trained with the Adam optimizer at a fixed initial learning rate of 10^{-4} and weight decay 10^{-4}. The learning rate is reduced by a factor of 0.5 when validation loss has not improved for 25 consecutive epochs (`ReduceLROnPlateau`), with a minimum floor of 10^{-5}. For every fold, the checkpoint with the highest validation accuracy is saved and used for MC inference. Training is performed on Kaggle using a Tesla T4 GPU with Automatic Mixed Precision (AMP) enabled.

Table 2 summarizes the per-experiment hyperparameters. The paper-based audio model uses a lower dropout rate (0.2) to match Spata et al.; all other models use 0.4. Video and multimodal models use a smaller batch size (16) reflecting the higher per-sample memory cost of the frame tensor compared with a single spectrogram.

Table 2: Hyperparameter summary per experiment.

Experiment	Arch.	Epochs	Batch	Dropout	Params	Frames
Audio Lightweight	GAP	200	32	0.4	153K	—
Audio Paper (CNNMC)	Spata	100	32	0.2	25M	—
Video Lightweight	GAP	200	16	0.4	153K	16
Multimodal Fusion	GAP+GAP	200	16	0.4	307K	16

8 Results

8.1 Summary Metrics

Table 3 reports the mean and standard deviation of MC accuracy and Equal Error Rate (EER) across the five cross-validation folds for all four experimental configurations.

Table 3: Final performance across the four experiments (mean ± std over 5 folds).

Experiment	Params	MC Accuracy	EER	Δ vs. Audio LW
Audio Lightweight	153K	0.285 ± 0.040	0.106 ± 0.014	—
Audio Paper (CNNMC)	25M	0.818 ± 0.025	0.016 ± 0.006	+53.3 pp
Video Lightweight	153K	0.875 ± 0.022	0.004 ± 0.002	+59.0 pp
Multimodal Fusion	307K	$\mathbf{0.940 \pm 0.029}$	$\mathbf{0.0019 \pm 0.0013}$	+65.5 pp

Efficiency paradox. The most striking finding is that the Video Lightweight model (153K parameters) outperforms the Audio Paper model (25M parameters) by +5.7 percentage points in MC accuracy while achieving a 9.7× lower EER. This holds despite the paper model having 163 times more parameters. The result shows that *modality quality can dominate model capacity*: when the input signal is intrinsically more discriminative, even a compact architecture can surpass a much larger model operating on a harder modality.

8.2 Per-Fold Breakdown

Tables 4 and 5 report the individual fold results for the audio experiments and for the video/multimodal experiments respectively.

Table 4: Fold-level results: Audio experiments.

	Audio Lightweight		Audio Paper (CNNMC)	
Fold	MC Acc	EER	MC Acc	EER
1	0.285	0.122	0.785	0.027
2	0.215	0.115	0.815	0.015
3	0.323	0.105	0.846	0.010
4	0.277	0.107	0.800	0.011
5	0.323	0.082	0.846	0.013
Mean	0.285	0.106	0.818	0.016
Std	0.040	0.014	0.025	0.006

Table 5: Fold-level results: Video and Multimodal experiments.

	Video Lightweight		Multimodal Fusion	
Fold	MC Acc	EER	MC Acc	EER
1	0.892	0.00256	0.954	0.00082
2	0.892	0.00292	0.962	0.00130
3	0.892	0.00192	0.969	0.00082
4	0.862	0.00457	0.892	0.00220
5	0.838	0.00776	0.923	0.00437
Mean	0.875	0.00395	0.940	0.00190
Std	0.022	0.00210	0.029	0.00133

8.3 Training Curves

Figure 1 shows the training and validation loss and accuracy for the Audio Paper (CNNMC) model on Fold 2. The curves exhibit the textbook signature of *overfitting*: training accuracy converges to near-perfect while validation accuracy plateaus around 75%, and the train-validation loss gap widens continuously after approximately epoch 20. This behavior is expected given that 25M parameters are optimized on only ∼520 training samples (roughly 12 per speaker). SpecAugment provides regularization but cannot compensate for a capacity-to-data ratio of ∼48 000 parameters per training sample.

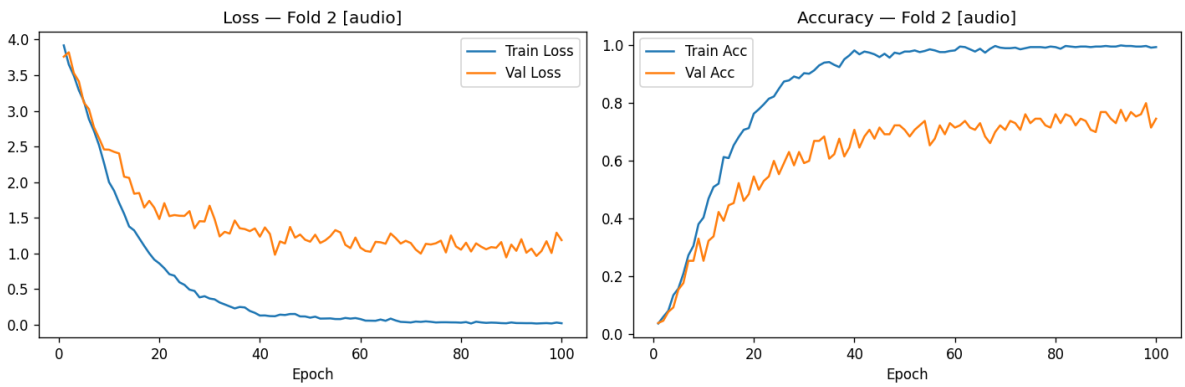

Figure 1: Loss and accuracy curves for the Audio Paper (CNNMC) model, Fold 2. Textbook overfitting: training accuracy reaches ∼100% while validation accuracy plateaus at ∼75%. The diverging loss gap opens at epoch 20–25.

Figure 2 shows the corresponding curves for the Multimodal Fusion model on Fold 3. In contrast to the audio paper experiment, training and validation metrics evolve together throughout the 200-epoch run, indicating that the lightweight architecture generalizes well. The validation oscillations visible in the accuracy panel reflect the small batch size (16) producing noisier gradient estimates; this is a known trade-off between gradient diversity and curve smoothness that in this case favours generalization.

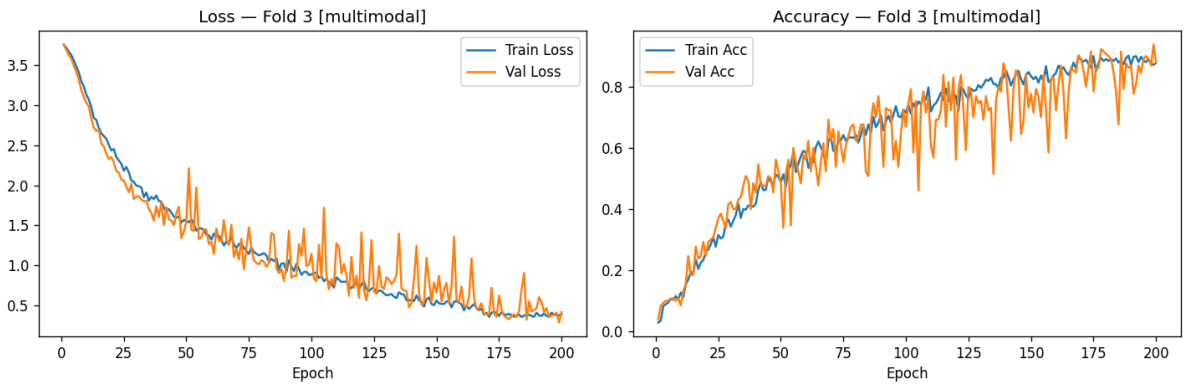

Figure 2: Loss and accuracy curves for the Multimodal Fusion model, Fold 3. Training and validation improve together throughout the run with no significant generalization gap, confirming that the lightweight fusion architecture does not overfit despite 200 epochs of training.

9 Dashboard Analysis

An interactive Streamlit dashboard was developed to visualize and explore the experimental results across all four configurations. The following subsections describe each dashboard panel and the analytical insights it conveys.

Note to authors: the figures below require screenshots captured directly from the running Streamlit application (`streamlit run dashboard/overview_dashboard.py`). Replace each placeholder with the corresponding screenshot before final submission.

9.1 Overview Dashboard

Figure 3: Overview dashboard comparing MC accuracy and EER for all four experimental configurations side by side.

The overview panel provides an immediate global comparison of the four systems. A grouped bar chart of MC accuracy and EER shows the full performance hierarchy: Audio Lightweight $\ll$ Audio Paper $<$ Video Lightweight $<$ Multimodal Fusion. The panel also renders the mean

uncertainty ($\bar{\sigma}$) per experiment, confirming that high accuracy correlates with low uncertainty in this study.

9.2 Cross-Validation Dashboard

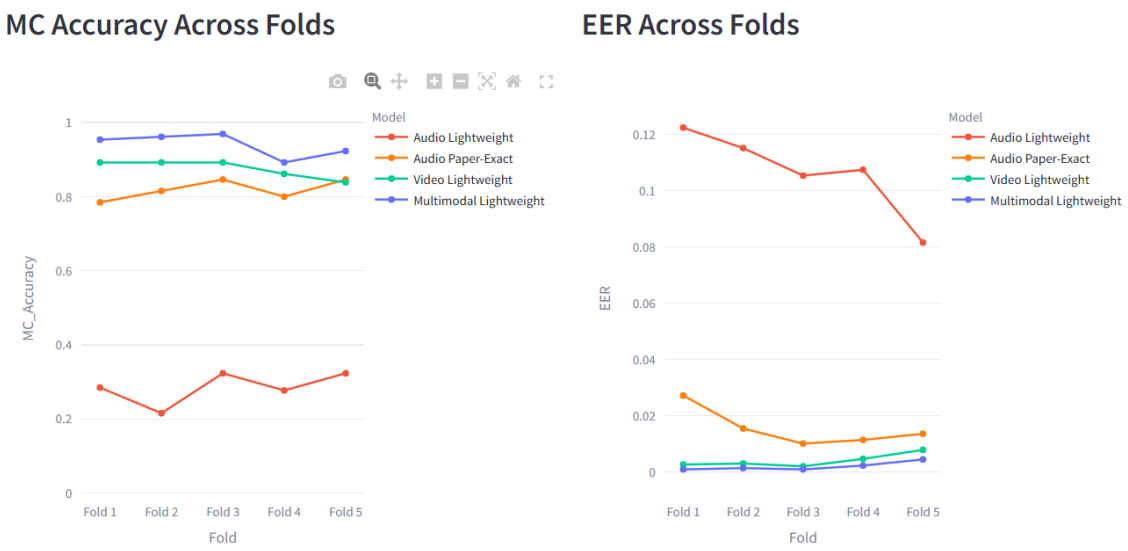

Figure 4: Cross-validation dashboard showing per-fold MC accuracy and EER for all four experiments.

The fold-by-fold panel addresses the robustness of the results. The ranking between models is consistent across all five folds, confirming that the observed hierarchy does not arise from a favorable data partition. Fold 4 is notable for showing slightly weaker performance in the multimodal system (MC Acc = 0.892 vs. the mean of 0.940), reflecting natural variance in small datasets; however, even this weakest fold outperforms the audio paper architecture in terms of EER.

9.3 Uncertainty and MC Dropout Dashboard

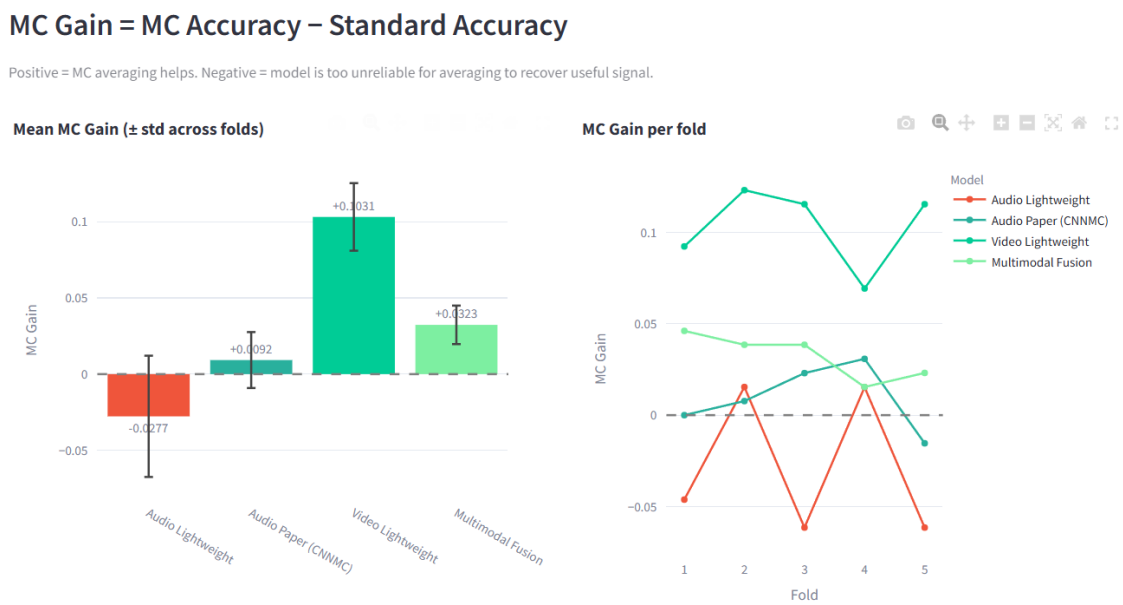

Figure 5: Uncertainty dashboard: standard vs. MC accuracy per experiment, and mean predictive uncertainty per model.

The uncertainty panel shows the accuracy gain from MC inference over single-pass standard inference, along with per-model mean uncertainty $\bar{\sigma}$. The Video Lightweight model shows the largest absolute gain from averaging ($\sim +9.5$ pp), consistent with the noisy single-pass predictions produced by smaller batch training. The multimodal model exhibits both the highest accuracy and the lowest uncertainty, confirming that it is not only the best-performing but also the most reliable system. The Audio Paper model changes the least from standard to MC inference, suggesting its larger capacity already produces relatively stable single-pass predictions.

10 Discussion

Modality quality dominates model capacity. The most important finding of this study is that the Video Lightweight model (153K parameters) surpasses the Audio Paper model (25M parameters) by 5.7 percentage points in MC accuracy, while achieving an EER nearly 10× lower. This result challenges the intuition that larger models always yield better performance and underscores that the discriminative richness of the input representation can matter more than the expressive power of the network. Lip movements during speech appear to contain highly reliable speaker-specific patterns in the BioVid recordings, which a small convolutional encoder can exploit directly.

Audio requires greater capacity. The dramatic improvement from 28.5% (Audio LW) to 81.8% (Audio Paper) when capacity is increased 163-fold confirms that audio does carry strong identity information in BioVid, but that this information is encoded in subtle spectral patterns that require a deeper and wider network to model. SpecAugment provides useful regularization for the paper model but cannot prevent overfitting entirely when 25 million parameters are fitted

on ∼520 training samples. Future work could explore early stopping, stronger augmentation, or a medium-capacity architecture between the two extremes.

Fusion yields complementary gains. The multimodal model adds 6.5 percentage points of MC accuracy over the video model alone, and reduces EER from 0.39% to 0.19%. Although the video stream is already highly informative, the audio branch provides complementary cues—particularly for speakers whose lip-movement patterns may be visually similar. The architecture cost of fusion is minimal: 307K vs. 153K parameters.

Calibration and uncertainty. The MC inference results confirm that the strongest models are also the most calibrated. The multimodal model achieves the highest accuracy and the lowest mean uncertainty, making it the most suitable system for practical biometric deployment, where abstention or escalation based on confidence level is often required.

11 Conclusion

This report extended the CNNMC framework of Spata et al. [1] to the visual and multimodal domains on the BioVid dataset. Four experimental configurations were evaluated under a shared 5-fold cross-validation protocol with Monte Carlo Dropout inference ($T = 30$).

The key results are as follows. A lightweight GAP architecture is insufficient to extract reliable identity information from audio alone (28.5% MC accuracy). Replicating the paper's architecture on audio raises performance to 81.8%, closely approaching the 93.27% reported in the original work. A lightweight GAP architecture applied to lip-video frames achieves 87.5% MC accuracy, surpassing the much larger audio model and establishing an efficiency paradox: 153K-parameter video model > 25M-parameter audio model. The multimodal late-fusion model achieves the best overall result: 94.0% MC accuracy and 0.19% EER, combining the two modalities with a total parameter budget of just 307K.

Methodologically, the project highlights that performance in deep learning depends not only on model capacity but critically on input modality quality and the interaction between the two. The MC Dropout framework of Spata et al. extends naturally to both visual and multimodal inputs, providing uncertainty estimates that add practical value to the identification pipeline.

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